

Reference papers 1

Latency, token use, cost, quality and limitations

12 papers in dama-archival/referencepapers_1 | 2 October 2026

Purpose: identify what existing work demonstrates, where its measurements stop, and what DAMA should test next. All performance figures are author-reported unless explicitly marked derived. No experiments were run for these reports.

Reading rule	Meaning
Compare within an experiment	A baseline and method in the same row share the source experiment. Scores across papers are not a common leaderboard, even when both say LoCoMo.
Timing has a scope	Write decision, ingestion, retrieval, complete answer, batch runtime and batch-amortized seconds/query are different measurements. p50 is median; p95 is tail latency.
Tokens are not automatically money	Prompt tokens, generated tokens, memory-injected context, stored representation and total lifecycle tokens differ. Actual prices and read/write frequency determine cost.
NR / N/A / derived	NR: a numerical result was not reported or could not be established in the reviewed source. N/A: no original system result, e.g. a survey. Derived reduction = $100 \times (\text{baseline} - \text{method}) / \text{baseline}$.
Quality and evidence	pp means percentage points. F1, judge score, task reward and human believability are distinct. Uncertainty is preserved where reported; no invented confidence intervals.

Evidence method: online primary papers, publisher records, author-posted text and official repositories were researched, then numerical tables were checked against the supplied PDFs. Image-only or malformed tables were visually inspected. Access limits and conflicting statements are noted on each evidence page.

Scope: principal reported efficiency, quality, resource and relevant ablation results for every paper in the selected folder. Hyperparameters and unrelated results from papers cited by surveys are not exhaustively reproduced. Proposed DAMA objectives are research targets, not findings.

COMPARISON MATRIX

Reported outcomes / 1-6

Each row keeps its own baseline and workload. See the evidence page for exact numbers, source location and qualification. NR does not mean zero overhead.

Paper	Latency	Token efficiency	Cost	Quality / outcome	Short limitation
R01 MemGPT	NR	NR	NR	DMR 93.4% vs 35.3%	Memory control requires LLM generation and tool calls.
R02 Generative Agents	Multiple days / simulation	NR	Thousands of USD; no saving %	Human-rated believability	Short simulated duration, domain-specific behavior evaluation, and expensive reflection/planning.
R03 LongLLMLingua	1.4-2.6x end-to-end speedup	82.3% fewer (LongBench)	52.6-94.0% API estimates	LongBench 48.3 vs 44.0	Compression can discard evidence and adds local computation.
R04 SwiftSage	NR in seconds	59.2-74.6% fewer tokens/action	No measured USD saving	ScienceWorld score 84.68	Fast/slow action planning, not persistent memory control.
R05 LongMemEval	No comparable system latency	115k / 1.5M-token test histories	N/A benchmark	500 questions; 5 capabilities	Answer quality alone cannot establish cost or forgetting safety.
R06 Zep	89.8-91.1% lower answer latency	98.6% fewer context tokens	NR monetary saving	LME S: 71.2% vs 60.2%	Graph ingestion uses LLM work outside the answer comparison.

COMPARISON MATRIX

Reported outcomes / 7-12

Each row keeps its own baseline and workload. See the evidence page for exact numbers, source location and qualification. NR does not mean zero overhead.

Paper	Latency	Token efficiency	Cost	Quality / outcome	Short limitation
R07 A-MEM	5.4 s API / 1.1 s local memory processing	85-93% fewer context tokens	<\$0.0003 per operation claim	Temporal F1 45.85 vs 25.52	Multiple LLM calls are used for notes and evolution.
R08 Mem0	p95: 17.117 -> 1.440 s	93.2% fewer (Mem0)	>90% token-cost headline	Judge 66.88 vs full 72.90	Full context still scores higher in this setup.
R09 Memory survey	N/A survey	N/A survey	N/A survey	Taxonomy and evaluation resources	A research map cannot prove the novelty or performance of DAMA.
R10 MemRouter	Write p50: 970 -> 58 ms	~38% turns omitted; tokens NR	No measured USD reduction	Matched F1: 52.0 vs 45.6	Learns ADD/NOOP admission, not the full lifecycle.
R11 SAGE	Ingestion 2.5x faster; total 23.3% lower	61.2% fewer write tokens	71.0% lower add API cost	Mean judge -1.3 pp vs Mem0	Novelty is not future usefulness.
R12 Gated-Memory Routing	HE: 3.0 -> 1.5 s amortized/query	Raw token saving NR	31.9% lower compute-cost proxy	Avg accuracy +2.44 pp	Memory is within a task, not persistent cross-session personalization.

MemGPT: Towards LLMs as Operating Systems

Memory paging controlled by the answer LLM; multi-session conversational recall and document analysis. DMR baselines use the corresponding fixed-context model.

Metric / scope	Baseline	Reported method result	Interpretation / change
DMR accuracy, GPT-3.5 Turbo	38.7%	66.9%	+28.2 percentage points (pp), derived
DMR accuracy, GPT-4	32.1%	92.5%	+60.4 pp, derived
DMR accuracy, GPT-4 Turbo	35.3%	93.4%	+58.1 pp, derived
ROUGE-L recall, GPT-4 Turbo	0.359	0.827	+0.468, derived
Latency / throughput	NR	NR	No quantitative latency reduction established
Token or monetary saving	NR	NR	Fixed context and paging are architectural mechanisms, not measured savings

Limitations. Memory control requires LLM generation and tool calls. DMR has short histories by modern standards; the large gain is against the paper baseline, not every possible full-history implementation.

For DAMA. Use as a functional memory baseline. Test whether a separate controller lowers total overhead while preserving recall.

Source: [R01] Table 2 (DMR), Table 3 (conversation openers), Sections 2 and 3. Online primary full text and supplied PDF reviewed.

Generative Agents: Interactive Simulacra of Human Behavior

ChatGPT-based agents combine memory streams, retrieval, reflection, and planning. A Smallville simulation studies 25 agents over two simulated days; 100 human evaluators rank behavioral believability.

Metric / scope	Baseline	Reported method result	Interpretation / change
Simulation scope	Ablations and human-written responses	25 agents; 2 simulated days	Behavioral study, not a memory-efficiency benchmark
Quality	Ablated architectures	Full architecture ranked most believable	Human ranking / TrueSkill; no QA accuracy equivalence
Runtime	No matched timing baseline	Multiple real-world days	Whole simulation, not seconds per answer
API expenditure	No saving baseline	Thousands of dollars in token credits	Approximate absolute statement; no exact bill
Token count / saving	NR	NR	No measured compression or cost-reduction percentage

Limitations. Short simulated duration, domain-specific behavior evaluation, and expensive reflection/planning. Believability does not establish factual memory reliability.

For DAMA. Borrow episodic storage and selective reflection; measure their cost and whether reflection improves future tasks.

Source: [R02] Sections 6 (evaluation), 7 (simulation), and 8.1 (limitations). Online primary full text and supplied PDF reviewed.

LongLLMLingua: Accelerating and Enhancing LLMs in Long Context Scenarios via Prompt Compression

Question-aware prompt compression; GPT-3.5-Turbo answer model, V100-32G for compression timing. End-to-end timing includes compression plus API request time.

Metric / scope	Baseline	Reported method result	Interpretation / change
LongBench prompt tokens	10,295	1,822	82.3% fewer; derived; about 6x compression
LongBench average score	44.0	48.3	+4.3 points, derived
LongBench latency	15.6 s	6.1 s	60.9% lower; derived; about 2.6x speedup
API USD / 1,000: multi-doc QA	\$4.6	\$1.3	71.7% lower; author estimate
API USD / 1,000: LongBench	\$31.5	\$3.0	90.5% lower; author estimate
API USD / 1,000: ZeroSCROLLS	\$30.6	\$3.2	89.5% lower; author estimate
API USD / 1,000: MuSiQue	\$3.8	\$1.8	52.6% lower; author estimate
API USD / 1,000: LooGLE	\$93.6	\$5.6	94.0% lower; author estimate

Limitations. Compression can discard evidence and adds local computation. It operates on available prompts; it does not maintain persistent memories or handle fact updates.

For DAMA. Use as an optional context-packet compressor and a strong compression baseline.

Table 9 uses historical API pricing and includes input/output inference estimates, not a current invoice or full compression-infrastructure cost.

Source: [R03] Tables 1-2; Appendix Table 9; latency evaluation; limitations. Online primary full text and supplied PDF reviewed.

SwiftSage: A Generative Agent with Fast and Slow Thinking for Complex Interactive Tasks

ScienceWorld: 30 task types, 270 test variations. A fine-tuned T5-large (770M) handles frequent actions; GPT-4 plans when needed.

Metric / scope	Baseline	Reported method result	Interpretation / change
ScienceWorld task score	ReAct 36.43; Reflexion 45.34	84.68	Task reward score, not memory-QA accuracy
Tokens per action vs SayCan	1,855.84	757.07	59.2% fewer, derived
Tokens per action vs ReAct	1,971.03	757.07	61.6% fewer, derived
Tokens per action vs Reflexion	2,983.46	757.07	74.6% fewer, derived
Scores per action	ReAct 0.79; Reflexion 0.13	2.73	Appendix aggregate efficiency measure
Measured seconds / USD saving	NR	NR	Action counts and token efficiency do not establish wall-clock speed

Limitations. Fast/slow action planning, not persistent memory control. It needs task demonstrations and uses rule-based switching.

For DAMA. Supports a small-model fast path with occasional larger-model escalation; DAMA must independently validate memory decisions.

Source: [R04] Table 1; Section 4.3; Appendix C.3 / Table 4. Online primary full text and supplied PDF reviewed.

LongMemEval: Benchmarking Chat Assistants on Long-Term Interactive Memory

Evaluation resource with 500 questions covering extraction, multi-session reasoning, temporal reasoning, knowledge updates, and abstention.

Metric / scope	Baseline	Reported method result	Interpretation / change
History size	Oracle evidence-only setting	S: about 115k; M: about 1.5M tokens	Dataset size, not tokens saved by a system
Long-context quality degradation	Oracle evidence-only reading	30-60% decline discussed in paper	Varies by reader and experimental setting
Fact-augmented retrieval keys	Unexpanded keys	9.4% higher recall@k; 5.4% higher QA accuracy	Authors' summary terminology; not a general efficiency claim
Temporal query expansion	Time-agnostic retrieval	6.8-11.3% recall improvement	Depends on expansion model
Structured reading / Chain-of-Note	Baseline reading	Up to 10 absolute accuracy points	Across three LLMs
Latency, USD, token reduction	N/A as a benchmark	No single system-wide result	Use to measure DAMA; do not rank as a deployed memory system

Limitations. Answer quality alone cannot establish cost or forgetting safety. Full ingestion accounting and controller-specific metrics must be added.

For DAMA. Use the full evaluation, including abstention and updates; keep train/dev/test histories separate.

Source: [R05] Sections 3-5; Table 1; indexing, query-expansion and reading experiments. Online primary full text and supplied PDF reviewed.

Zep: A Temporal Knowledge Graph Architecture for Agent Memory

LongMemEval S, GPT-4o and GPT-4o-mini. Paper reports a latency statistic with IQR. Zep was accessed over a network; baseline did not have the same network path.

Metric / scope	Baseline	Reported method result	Interpretation / change
Accuracy, GPT-4o-mini	55.4%	63.8%	+8.4 pp; +15.2% relative, derived
Latency, GPT-4o-mini	31.3 s (IQR 8.76)	3.20 s (IQR 1.31)	89.8% lower, derived
Accuracy, GPT-4o	60.2%	71.2%	+11.0 pp; +18.3% relative from rounded values
Latency, GPT-4o	28.9 s (IQR 6.01)	2.58 s (IQR 0.684)	91.1% lower, derived
Context per question	115k tokens	1.6k tokens	98.6% fewer, derived
DMR, GPT-4 Turbo	Full conversation 94.4%	94.8%	+0.4 pp; MemGPT cited result 93.4%
Monetary saving	NR	NR	Graph construction and maintenance must also be costed

Limitations. Graph ingestion uses LLM work outside the answer comparison. Some categories regress with the smaller model. Results are author-reported and deployment-specific.

For DAMA. Use temporal validity and provenance; measure graph build costs as well as retrieval.

The abstract says up to 18.5% accuracy improvement; rounded Table 2 values yield 18.3%. This report retains raw values and labels the calculation.

Source: [R06] Tables 1-3; Sections 4.2-4.3 and limitations. Online primary full text and supplied PDF reviewed.

A-MEM: Agentic Memory for LLM Agents

LoCoMo across six answer models; linked notes with LLM-driven memory evolution. Context tokens in Table 1 and memory-processing time are different scopes.

Metric / scope	Baseline	Reported method result	Interpretation / change
QA prompt tokens, GPT-4o-mini	MemGPT 16,977	2,520	85.2% fewer, derived
QA prompt tokens, GPT-4o	MemGPT 16,987	1,216	92.8% fewer, derived
Temporal F1, GPT-4o-mini	MemGPT 25.52	45.85	+20.33 points, derived; other categories vary
Memory processing time	No paired timing baseline	5.4 s (GPT-4o-mini); 1.1 s (local Llama 3.2 1B)	Different models/deployments; not a controlled speedup
Operation cost	NR baseline	Less than \$0.0003 / memory operation	Authors' estimate; contemporary prices may differ
Storage at 1k / 1M memories	NR total-system baseline	1.46 / 1,464.84 MB	Index memory; not model RAM
Retrieval at 1k / 1M memories	See source Table 4	0.31 / 3.70 microseconds as printed	Reported units need reproduction; not end-to-end answer time

Limitations. Multiple LLM calls are used for notes and evolution. No complete learned lifecycle controller; deletion safety is not established. Table 4 microsecond units should not be converted into an end-to-end claim.

For DAMA. Compare adaptive organization with simpler storage at an equal context budget; charge every evolution call.

Source: [R07] Tables 1 and 4; Section 4.3 cost-efficiency analysis. Online primary full text and supplied PDF reviewed.

Mem0: Building Production-Ready AI Agents with Scalable Long-Term Memory

LoCoMo; GPT-4o-mini generation setup. Mem0 and graph-enhanced Mem0g are compared with full context and other systems within the paper's harness.

Metric / scope	Baseline	Reported method result	Interpretation / change
Answer latency p50	Full context 9.870 s	Mem0 0.708 s; Mem0g 1.091 s	92.8% / 88.9% lower, derived
Answer latency p95	Full context 17.117 s	Mem0 1.440 s; Mem0g 2.590 s	91.6% / 84.9% lower, derived
Search p50 / p95	Separate from generation	Mem0 0.148 / 0.200 s; Mem0g 0.476 / 0.657 s	Retrieval-only timing
Memory/context tokens in Table 2	26,031	Mem0 1,764; Mem0g 3,616	93.2% / 86.1% fewer, derived
Overall judge score	Full context 72.90; OpenAI baseline 52.90	Mem0 66.88; Mem0g 68.44	vs full: -6.02 / -4.46 pp; vs OpenAI: +26.4% / +29.4% relative
Stored representation per conversation	Raw dialogue about 26k tokens	Mem0 about 7k; Mem0g about 14k	Storage representation, not per-question injected context
Monetary claim	Full-context inference	>90% token-cost saving headline	Not a measured all-inclusive deployment bill

Limitations. Full context still scores higher in this setup. Ingestion, memory updates, and graph maintenance are outside Table 2 answer timing. Current SDK/platform differs from this paper.

For DAMA. A priority baseline. DAMA should report total cost and a quality bound, not only its best token reduction.

Source: [R08] Table 2; Sections 4.3-4.5. Online primary full text and supplied PDF reviewed.

Memory in the Age of AI Agents: A Survey - Forms, Functions and Dynamics

Literature taxonomy and resource survey. It does not introduce one benchmarked system with a single latency/cost outcome.

Metric / scope	Baseline	Reported method result	Interpretation / change
Memory forms	N/A	Token-level, parametric, latent	Different representations have different resource costs
Functional roles	N/A	Factual, experiential, working memory	Useful dimensions for DAMA capability coverage
Lifecycle	N/A	Formation, retrieval, evolution	Includes consolidation and forgetting
Latency / tokens / monetary savings	N/A	No original single-system measurement	Results cited from other papers belong to their original experiments
Quality / storage score	N/A	No unified original benchmark score	Do not aggregate heterogeneous cited results into a ranking

Limitations. A research map cannot prove the novelty or performance of DAMA. Individual comparisons require original-paper evidence.

For DAMA. Use the taxonomy to define the implementation scope and ablations, and revisit relevant prior work before claiming novelty.

Source: [R09] Sections 3-7: forms, functions, dynamics, resources and frontiers. Online primary full text and supplied PDF reviewed.

MemRouter: Memory-as-Embedding Routing for Long-Term Conversational Agents

Matched LoCoMo harness: same Qwen2.5-7B answer model, retrieval, and prompts. Test set: 8 conversations, about 5,000 turns and 1,300 questions.

Metric / scope	Baseline	Reported method result	Interpretation / change
Overall QA F1	LLM manager 45.6	52.0	+6.4 points; non-overlapping 95% CIs reported
Memory decision p50 / p95	970 / 1,415 ms per turn	58 / 71 ms per turn	94.0% / 95.0% lower, derived
Total memory-management time	5,028 s	293 s	94.2% lower; 17.2x speedup, derived
QA latency p50	198 ms/question	141 ms/question	28.8% lower, derived
End-to-end throughput	0.24 QA/s	2.58 QA/s	10.75x, derived; not 17x
Peak GPU memory	26.6 GB	31.2 GB	17.3% higher, derived
Model / retention	Full LLM memory manager	About 12M trainable parameters; about 62% of turns stored	Frozen 7B backbone still required; retention is not token saving

Limitations. Learns ADD/NOOP admission, not the full lifecycle. Teacher labels and a frozen large backbone remain necessary; peak GPU memory increases.

For DAMA. Strong precedent for decoupling write decisions from answering. Aim for an actually deployable small controller and expand operations without losing recall.

Source: [R10] Section 4.3; Appendix B Tables 4-6; limitations. Online primary full text and supplied PDF reviewed.

SAGE: A Novelty Gate for Efficient Memory Evolution in Agentic LLMs

Full LoCoMo, GPT-4o-mini. A statistical novelty gate routes candidates to ADD, NOOP, or an LLM merge/update path. Separate A-MEM gate experiments use five backbones.

Metric / scope	Baseline	Reported method result	Interpretation / change
Add-phase API cost	Mem0 \$1.24	\$0.36	71.0% lower, derived; 3.4x cheaper
Add total tokens	5,551,631	2,155,207	61.2% fewer, derived
Add prompt / completion tokens	4,637,649 / 913,982	2,072,575 / 82,632	55.3% / 91.0% fewer, derived
Add call latency / ingestion time	5.38 s / 39.3 min	1.76 s / 15.7 min	67.3% / 60.1% lower, derived
Search / total wall-clock	66.7 / 106.3 min	65.5 / 81.5 min	1.8% / 23.3% lower, derived
Write/add LLM calls	3,217	2,297	28.6% fewer, derived
Quality trade-off (GPT-4o-mini)	Mem0 mean judge 53.5; temporal 42.7	Mean judge 52.2; temporal 35.4	-1.3 / -7.3 pp, derived
A-MEM gate transfer	Ungated A-MEM	15.8-17.9% calls skipped	GPT-4o-mini judge -2.01 pp; open-model effects vary

Limitations. Novelty is not future usefulness. It retains extraction costs and can harm temporal recall. This is primarily a write gate, not a complete memory controller.

For DAMA. Include a cheap novelty gate as a strong baseline; test usefulness and updates separately and preserve a fallback.

Source: [R11] Tables 1 and 4-5; write-side efficiency discussion. Online primary full text and supplied PDF reviewed.

Learning What to Retain: Gated-Memory Routing for Efficient Collaboration in Multi-Agent LLM Systems

Five reasoning/code benchmarks with a heterogeneous 3B-32B model pool. Write, retrieval, role/backbone routing and halting are learned jointly.

Metric / scope	Baseline	Reported method result	Interpretation / change
Average accuracy across 5 tasks	Puppeteer-32B 75.29%	77.73%	+2.44 pp
HumanEval pass@1	Puppeteer-32B 85.16%	89.84%	+4.68 pp, derived
HumanEval test-set cost proxy	Puppeteer 0.047; MASRouter 0.057	0.032	31.9% / 43.9% lower; size-weighted proxy, not USD
HumanEval batch-amortized time	Puppeteer 3.0 s; MASRouter 4.4 s	1.5 s/query	50.0% / 65.9% lower, derived
MBPP batch-amortized time	Puppeteer 3.2 s; MASRouter 6.4 s	2.6 s/query	18.8% / 59.4% lower, derived
Approximate PFLOPs/query	Puppeteer: HE 0.144; MBPP 0.163	HE 0.107; MBPP 0.145	25.7% / 11.0% lower, derived
HumanEval vs full-history routing	Cost 0.068; accuracy 89.06%	Cost 0.032; accuracy 89.84%	52.9% lower proxy cost; +0.78 pp
Raw total-token saving	NR paired aggregate	NR	Model-size-weighted cost cannot be relabeled as token reduction

Limitations. Memory is within a task, not persistent cross-session personalization. Batch-amortized time is not interactive p50/p95; proxy prices are deliberately synthetic.

For DAMA. Jointly evaluate writing, retrieval, and stopping. DAMA must additionally demonstrate persistence and temporal updates.

Source: [R12] Tables 1-3; Appendix 9 Tables 4-5; Appendix 10 Table 7. Online primary full text and supplied PDF reviewed.

WHAT THIS COLLECTION ESTABLISHES

Evidence to carry into DAMA

Finding	Evidence	What remains open
Compact context can work	LongLLMLingua, Zep, A-MEM and Mem0 report large context reductions in their own tests.	What happens when every write, retrieval, summary and fallback is charged to the same workload?
Write control can become cheaper	MemRouter removes per-turn decoding; SAGE avoids many LLM routing/merge calls.	Can a small deployable model handle admission, conflict updates and budgeting without losing rare evidence?
Quality is a constraint	Mem0 trails full-context quality in its table; SAGE loses temporal judge points in the GPT-4o-mini setup.	Savings should be accepted only at a declared quality tolerance, with category-level checks.
Small/large specialization is established	SwiftSage separates frequent actions from deliberate planning; MemRouter separates write decisions from answer generation.	DAMA needs a concrete policy or measured systems advantage beyond the presence of two agents.
A benchmark needs extensions	LongMemEval tests important memory abilities; the survey organizes the field.	Add full-lifecycle timing, token accounting, memory growth, harmful forgetting and calibrated fallback metrics.

Suggested first experiment

Keep the answer model fixed. Compare full context, a sliding window, vector retrieval, an LLM memory manager, Mem0, and DAMA V1 on the same history splits. Report a quality-versus-total-cost curve at multiple memory-packet budgets. Reproduce baseline behavior before training a specialized controller.

Paper results versus current software

Historical results should be reproduced with pinned code and configurations. Current software can implement a different algorithm, use newer models, or depend on managed services.

System	Online status checked	Implication for dama-compare
MemGPT / Letta	The original MemGPT project is now developed as Letta, a platform for stateful agents.	Treat original-paper MemGPT and a current Letta baseline as separate configurations. Record commit, agent type, model and memory settings.
Mem0 2025 paper	The 2025 paper reports LoCoMo answer timing and quality in its original setup.	Use these as historical claims, not promised performance of a new installation.
Mem0 current repository	README describes an April 2026 algorithm and managed-platform scores: LoCoMo 92.5, 7.0k tokens, p50 0.88 s; LongMemEval 94.4, 6.8k tokens, p50 1.09 s.	Vendor-reported, separate from the 2025 paper. README says proprietary platform optimizations are absent from the open-source SDK. Do not compare those scores directly with historical F1/judge values.
Local evaluation	Official Mem0 evaluation suite and LongMemEval code are available.	Open-source code does not imply fully local inference. Configure generation, embeddings and judge providers; include their resource use.

Official online sources

Letta - <https://github.com/letta-ai/letta>

Mem0 - <https://github.com/mem0ai/mem0>

Mem0 2025 paper - <https://arxiv.org/html/2504.19413>

Mem0 evaluation suite - <https://github.com/mem0ai/memory-benchmarks>

These repository statements were checked online on the research date. They have not been independently reproduced in this workspace.

Source register / 1-6

[R01] MemGPT: Towards LLMs as Operating Systems (2023; revised 2024)

[Online source](#) | Table 2 (DMR), Table 3 (conversation openers), Sections 2 and 3

Local source: dama-archival/referencepapers_1/01_MemGPT.pdf

Online primary full text and supplied PDF reviewed.

[R02] Generative Agents: Interactive Simulacra of Human Behavior (2023)

[Online source](#) | Sections 6 (evaluation), 7 (simulation), and 8.1 (limitations)

Local source: dama-archival/referencepapers_1/02_Generative_Agents.pdf

Online primary full text and supplied PDF reviewed.

[R03] LongLLMLingua: Accelerating and Enhancing LLMs in Long Context Scenarios via Prompt Compression (2023; ACL 2024)

[Online source](#) | Tables 1-2; Appendix Table 9; latency evaluation; limitations

Local source: dama-archival/referencepapers_1/03_LongLLMLingua.pdf

Online primary full text and supplied PDF reviewed.

[R04] SwiftSage: A Generative Agent with Fast and Slow Thinking for Complex Interactive Tasks (2023)

[Online source](#) | Table 1; Section 4.3; Appendix C.3 / Table 4

Local source: dama-archival/referencepapers_1/04_SwiftSage.pdf

Online primary full text and supplied PDF reviewed.

[R05] LongMemEval: Benchmarking Chat Assistants on Long-Term Interactive Memory (2024; ICLR 2025)

[Online source](#) | Sections 3-5; Table 1; indexing, query-expansion and reading experiments

Local source: dama-archival/referencepapers_1/05_LongMemEval.pdf

Online primary full text and supplied PDF reviewed.

[R06] Zep: A Temporal Knowledge Graph Architecture for Agent Memory (2025)

[Online source](#) | Tables 1-3; Sections 4.2-4.3 and limitations

Local source: dama-archival/referencepapers_1/06_Zep_Graphiti.pdf

Online primary full text and supplied PDF reviewed.

Online sources are clickable. Full URLs, raw comparison rows and local-file SHA-256 hashes are recorded in report_sources.json alongside these reports. Duplicate papers retain the same source IDs in both PDFs.

Source register / 7-12

[R07] A-MEM: Agentic Memory for LLM Agents (2025)

[Online source](#) | Tables 1 and 4; Section 4.3 cost-efficiency analysis

Local source: dama-archival/referencepapers_1/07_A-MEM.pdf

Online primary full text and supplied PDF reviewed.

[R08] Mem0: Building Production-Ready AI Agents with Scalable Long-Term Memory (2025)

[Online source](#) | Table 2; Sections 4.3-4.5

Local source: dama-archival/referencepapers_1/08_Mem0.pdf

Online primary full text and supplied PDF reviewed.

[R09] Memory in the Age of AI Agents: A Survey - Forms, Functions and Dynamics (2025; v2 Jan 2026)

[Online source](#) | Sections 3-7: forms, functions, dynamics, resources and frontiers

Local source: dama-archival/referencepapers_1/09_Memory_in_the_Age_of_AI_Agents_Survey.pdf

Online primary full text and supplied PDF reviewed.

[R10] MemRouter: Memory-as-Embedding Routing for Long-Term Conversational Agents (2026)

[Online source](#) | Section 4.3; Appendix B Tables 4-6; limitations

Local source: dama-archival/referencepapers_1/10_MemRouter.pdf

Online primary full text and supplied PDF reviewed.

[R11] SAGE: A Novelty Gate for Efficient Memory Evolution in Agentic LLMs (2026)

[Online source](#) | Tables 1 and 4-5; write-side efficiency discussion

Local source: dama-archival/referencepapers_1/11_SAGE_Novelty_Gate.pdf

Online primary full text and supplied PDF reviewed.

[R12] Learning What to Retain: Gated-Memory Routing for Efficient Collaboration in Multi-Agent LLM Systems (2026)

[Online source](#) | Tables 1-3; Appendix 9 Tables 4-5; Appendix 10 Table 7

Local source: dama-archival/referencepapers_1/12_Gated_Memory_Routing.pdf

Online primary full text and supplied PDF reviewed.

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